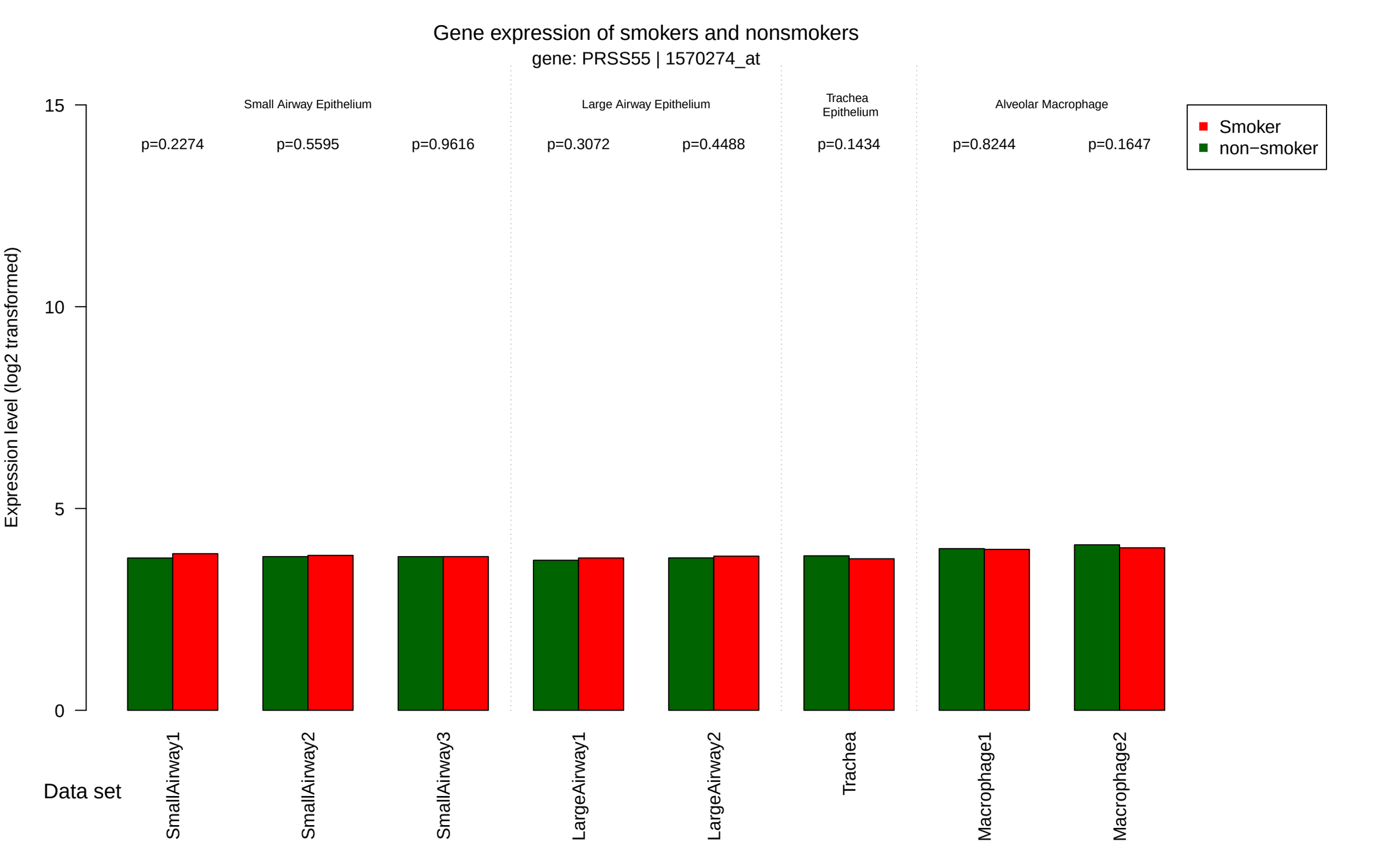
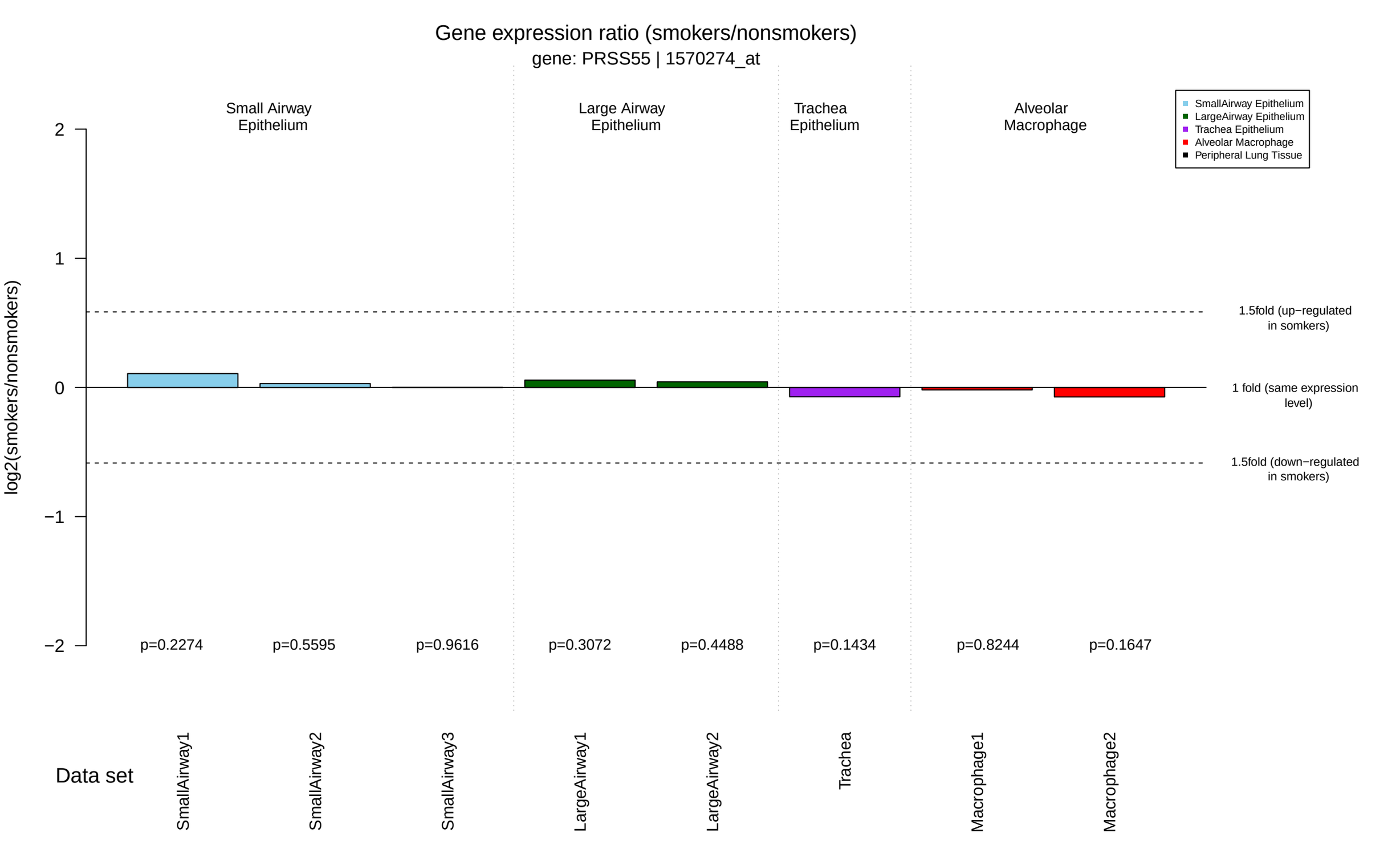
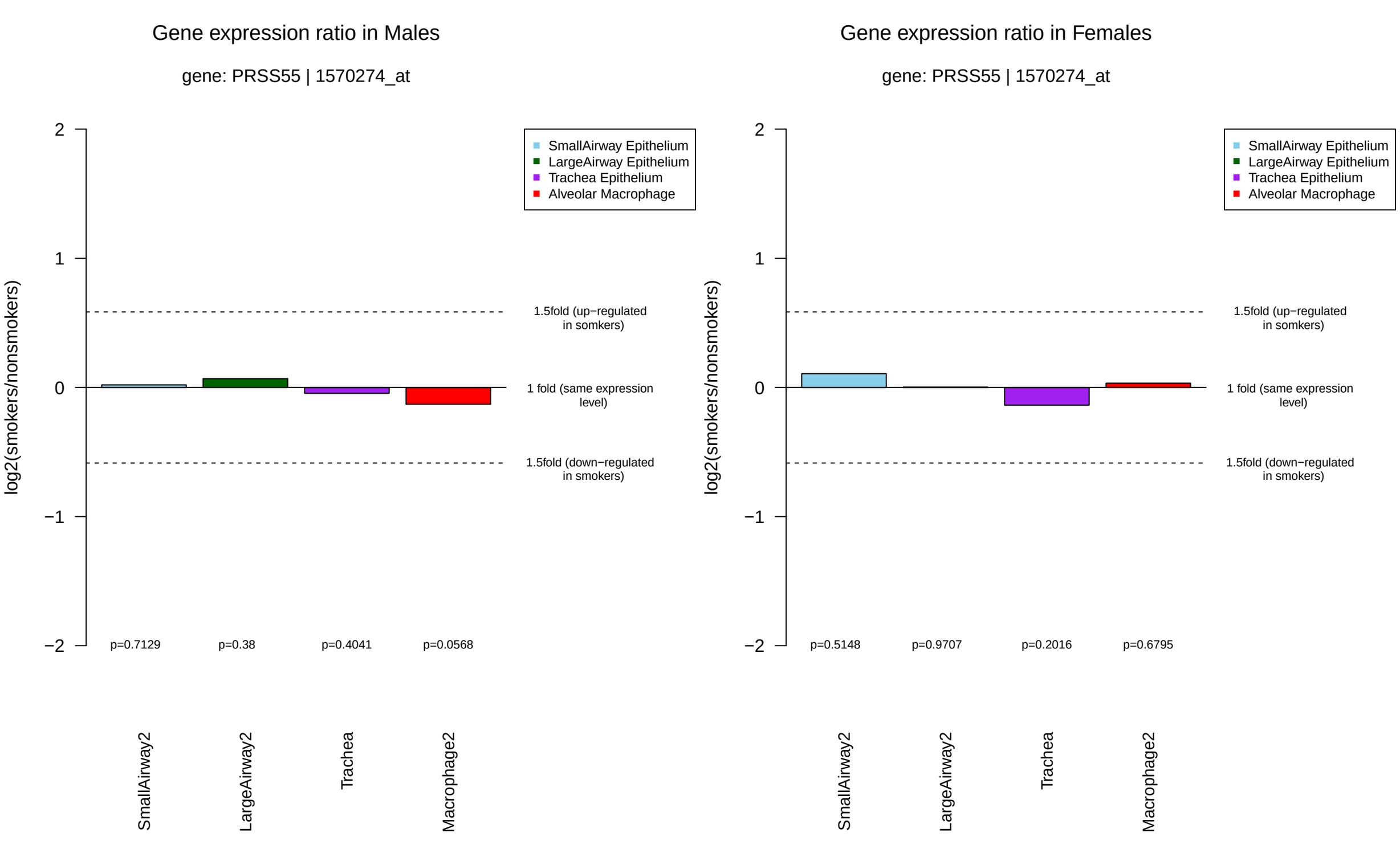
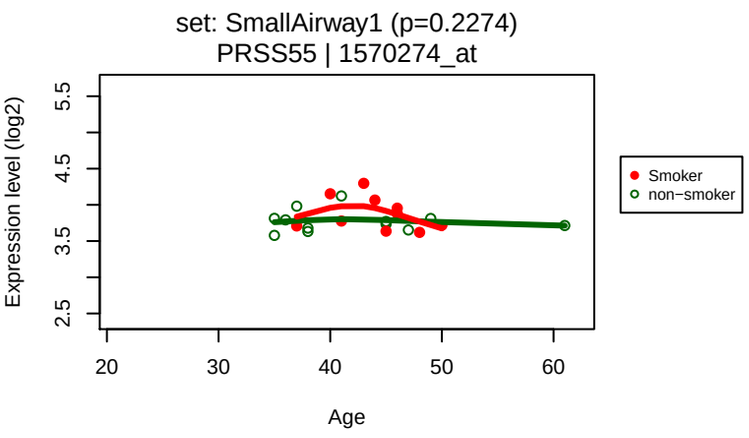
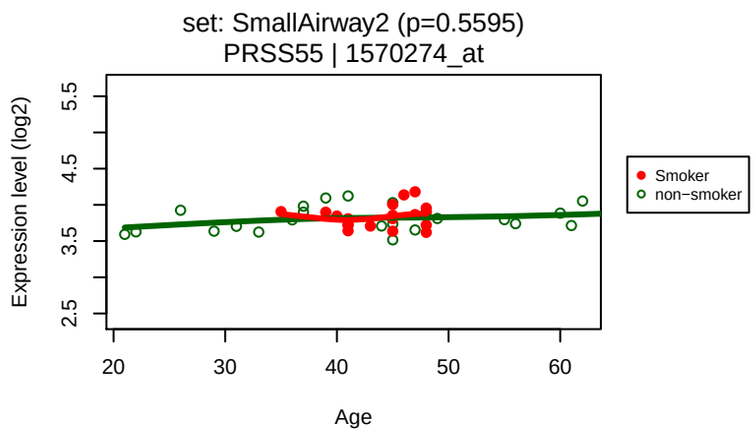
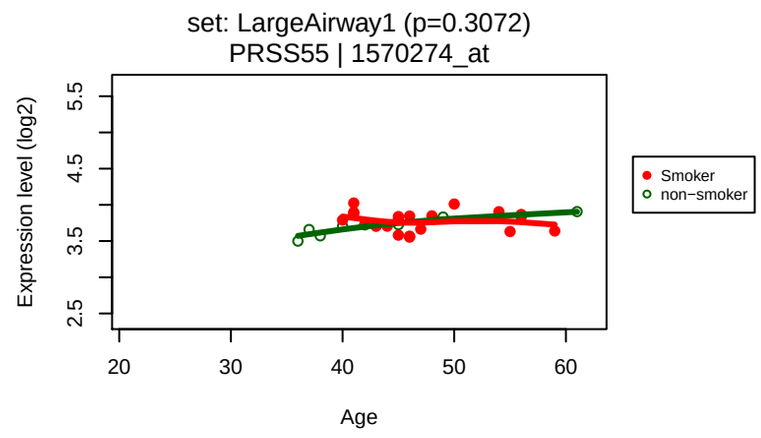

Gene expression of smokers and nonsmokers
gene: PRSS55 | 1570274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2274
p=0.5595
p=0.9616
p=0.3072
p=0.4488
p=0.1434
p=0.8244
p=0.1647
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

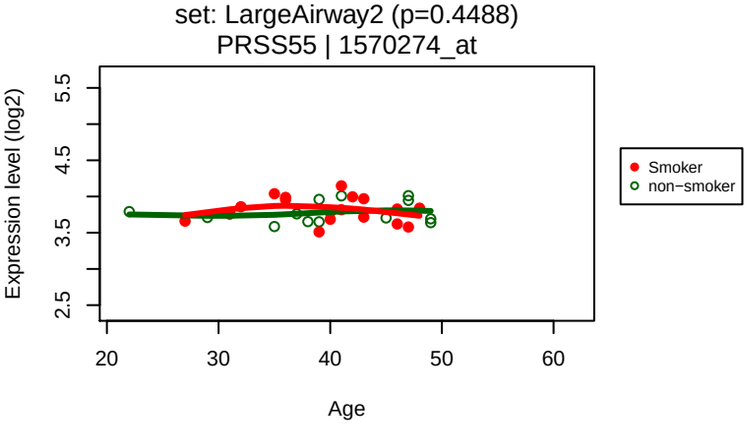

Gene expression ratio (smokers/nonsmokers)
gene: PRSS55 | 1570274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2274
p=0.5595
p=0.9616
p=0.3072
p=0.4488
p=0.1434
p=0.8244
p=0.1647
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

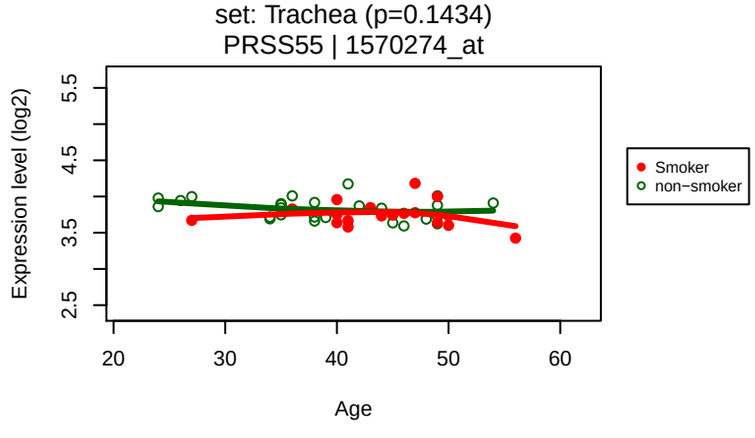

Gene expression ratio in Males
gene: PRSS55 | 1570274_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7129
p=0.38
p=0.4041
p=0.0568
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS55 | 1570274_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5148
p=0.9707
p=0.2016
p=0.6795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

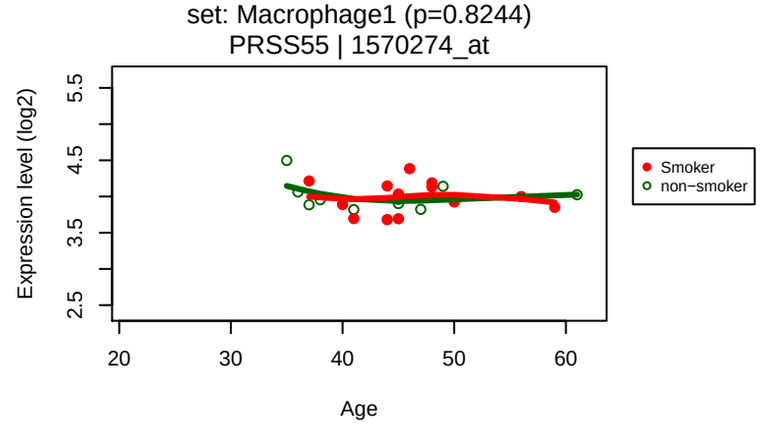

set: SmallAirway1 (p=0.2274)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker

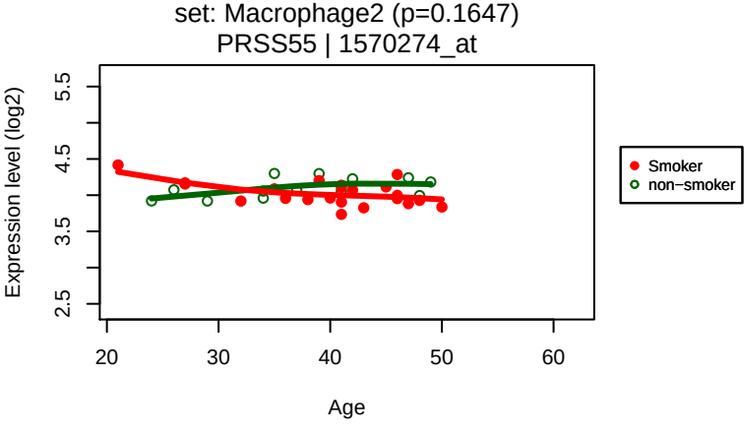

set: SmallAirway2 (p=0.5595)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker

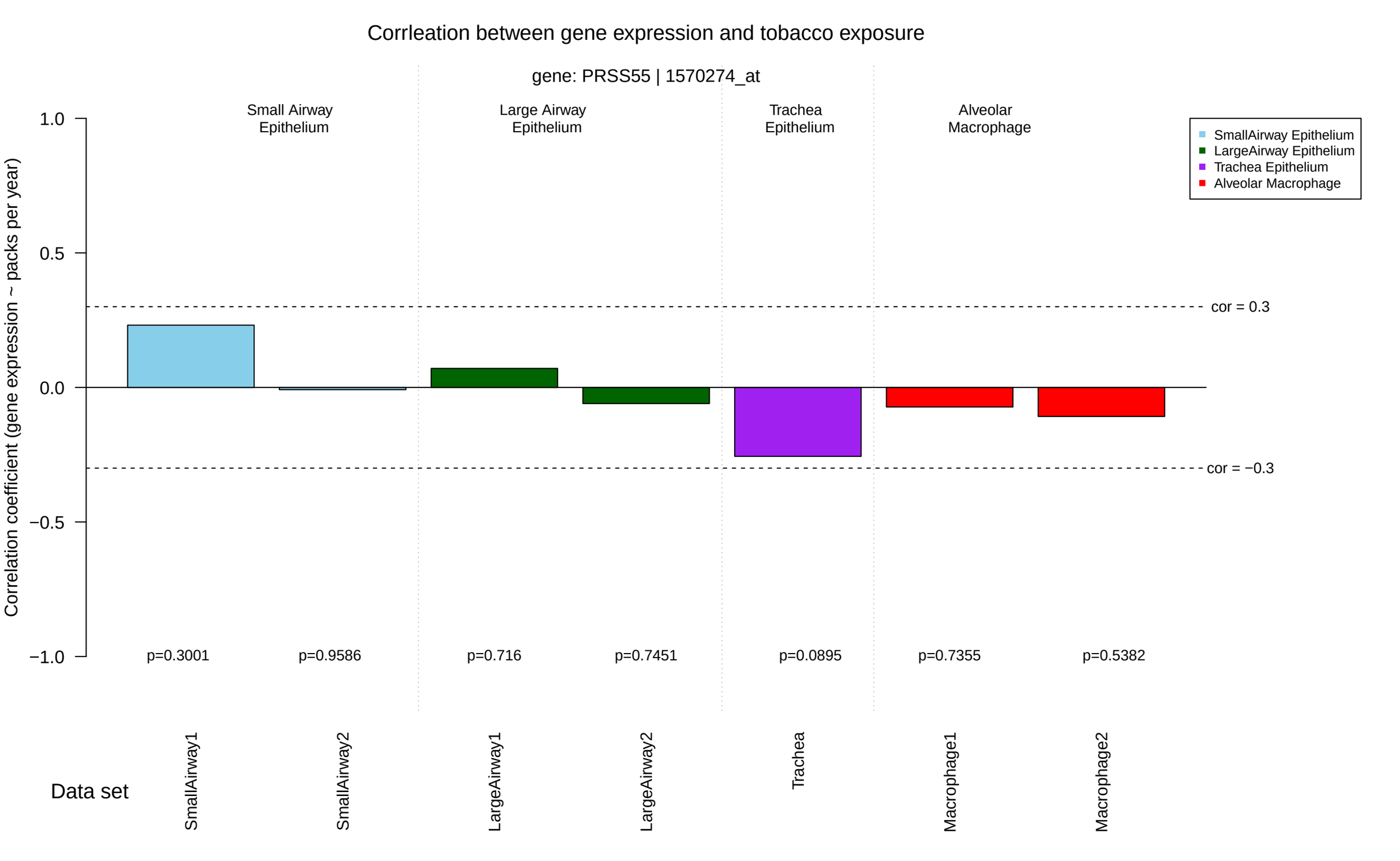

set: LargeAirway1 (p=0.3072)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4488)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1434)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8244)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1647)
PRSS55 | 1570274_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS55 | 1570274_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3001
p=0.9586
p=0.716
p=0.7451
p=0.0895
p=0.7355
p=0.5382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2